Assignment 2

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Econometrics I
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Instructions. This assignment covers Session 3 (inference and standard errors). You may use `R` (with `fixest`) or Python (with `pyfixest`). Turn in (1) a PDF presenting your results and showing your work, and (2) your code.

1 Monte Carlo: sampling distributions, standard errors, and coverage

Consider the data generating process

$$y_i = 1 + 2x_i + \varepsilon_i, \qquad x_i \sim \mathcal{N}(0, 1)$$

For each design below, simulate $S = 1{,}000$ datasets of size $n = 50$, estimate the regression by OLS in each, and save $\hat{\beta}_1^{(s)}$ and its reported standard error. Report: (i) the mean of $\hat{\beta}_1^{(s)}$; (ii) the standard deviation of $\hat{\beta}_1^{(s)}$ across simulations (the "true" standard error); (iii) the average reported standard error; and (iv) *coverage*: the fraction of simulations in which the 95% confidence interval contains $\beta_1 = 2$.

(a) $\varepsilon_i \sim \mathcal{N}(0, 4)$, classical standard errors. Does everything line up the way Sessions 1 and 3 promised?

(b) Heteroskedasticity: $\varepsilon_i = x_i^2 \cdot \nu_i$ with $\nu_i \sim \mathcal{N}(0, 1)$. Compute coverage using (i) classical and (ii) heteroskedasticity-robust (HC1) standard errors. Which one is lying to you, and in which direction?

(c) Clustering: assign observations to $G = 25$ groups of 20 (so $n = 500$ here), and let

$$\varepsilon_{ig} = \eta_g + \nu_{ig}, \qquad \eta_g \sim \mathcal{N}(0, 1), \quad \nu_{ig} \sim \mathcal{N}(0, 1)$$

with $x_{ig} = w_g + z_{ig}$ where $w_g, z_{ig} \sim \mathcal{N}(0, 1)$ (so the regressor is also correlated within group). Compute coverage using classical, HC1, and cluster-robust standard errors. Present the three coverage rates in one table and summarize the lesson in two sentences.

(d) In (c), how does cluster-robust coverage change if you re-run the design with $G = 5$ groups of 100? One sentence on what this says about "number of clusters" warnings you will encounter in applied work.

2 Bootstrap vs. the formulas

Return to design (a) of Question 1, but with a single simulated dataset of $n = 50$.

(a) Compute the standard error of $\hat{\beta}_1$ three ways: the classical formula, HC1, and a non-parametric bootstrap with $B = 2{,}000$ resamples (resample rows with replacement). Compare.

(b) Now compute all three for the *ratio* $\hat{\beta}_1/\hat{\beta}_0$. (For the formula-based versions, note there is no textbook formula — explain in one sentence why the bootstrap doesn't care.)

3 Fuel bills: the CEF in the wild

The file `fuelbills.csv` contains data on 144 homes: `FUELBILL` (annual fuel expenditure) and `ROOMS` (number of rooms), among other variables.

(a) Produce a scatterplot of `FUELBILL` against `ROOMS`. Compute and plot the mean of `FUELBILL` for each value of `ROOMS`. (You have just estimated a conditional expectation function.)

(b) Regress `FUELBILL` on `ROOMS` (with `feols` or `pf.feols`). Interpret the slope in one sentence. Overlay the fitted line on your plot from (a): where does the linear approximation fit the CEF well, and where does it not?

(c) Now run the *saturated* regression of `FUELBILL` on a full set of indicator variables for each value of `ROOMS` (in `fixest`: `feols(FUELBILL ~ i(ROOMS))`; in `pyfixest`: `pf.feols("FUELBILL ~ i(ROOMS)")`). Show that the fitted values reproduce the conditional means from (a) exactly. Why must this be true?

(d) Report the regression from (b) with classical and with heteroskedasticity-robust standard errors. Is there visual evidence in (a) that the robust ones are the right choice?